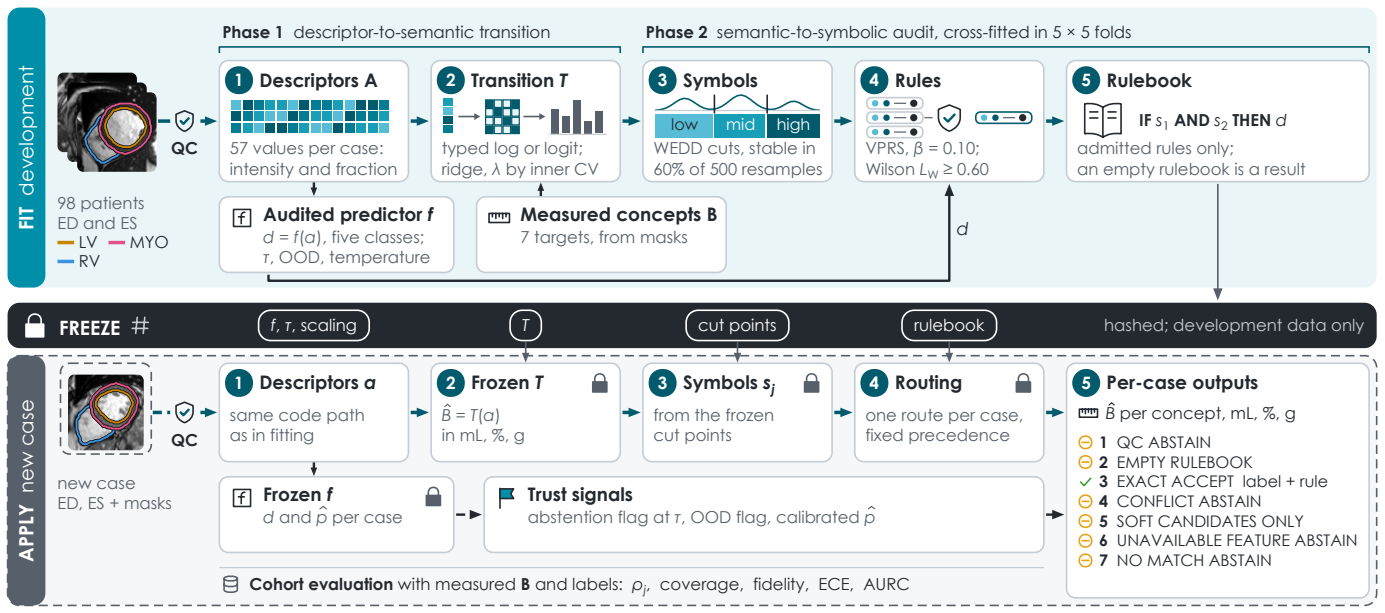

FIT development
Phase 1 descriptor-to-semantic transition
Phase 2 semantic-to-symbolic audit, cross-fitted in 5 × 5 folds
1 Descriptors A
57 values per case: intensity and fraction
2 Transition T
typed log or logit; ridge, λ by inner CV
3 Symbols
low mid high
WEDD cuts, stable in 60% of 500 resamples
4 Rules
VPRS, β = 0.10; Wilson L_W ≥ 0.60
5 Rulebook
IF s1 AND s2 THEN d
admitted rules only; an empty rulebook is a result
QC
98 patients
ED and ES
LV MYO
RV
Audited predictor f
d = f(a), five classes; τ, OOD, temperature
Measured concepts B
7 targets, from masks
d
FREEZE #
f, τ, scaling
T
cut points
rulebook
hashed; development data only
APPLY new case
1 Descriptors a
same code path as in fitting
2 Frozen T
B̂ = T(a) in mL, %, g
3 Symbols s_j
from the frozen cut points
4 Routing
one route per case, fixed precedence
5 Per-case outputs
B̂ per concept, mL, %, g
1 QC ABSTAIN
2 EMPTY RULEBOOK
3 EXACT ACCEPT label + rule
4 CONFLICT ABSTAIN
5 SOFT CANDIDATES ONLY
6 UNAVAILABLE FEATURE ABSTAIN
7 NO MATCH ABSTAIN
QC
new case
ED, ES + masks
Frozen f
d and p̂ per case
Trust signals
abstention flag at τ, OOD flag, calibrated p̂
Cohort evaluation with measured B and labels: ρ_j, coverage, fidelity, ECE, AURC